IE4483 – Artificial Intelligence and Data Mining **AY 2023 Semester 1**

Homework 5 **Due: Saturday, 11 November 2023**

1. Are the following statements true or false? Justify your answers.

   (a) K-Means clustering always generate the same result, for a given set of data samples.

   (b) HAC always generate the same result, for a given set of data samples.

   (c) Given a set of annotated data for training, increasing the model complexity will always lower the testing / validation error.

(60%)

2. You are to cluster 4 one-dimensional data samples, which are p1 = 1, p2 = 5, p3 = 6 and p4 = 8, into 2 clusters.

   (a) Show the HAC clustering process and results, with MIN distance.

   (b) Show the K-Means clustering process and results, using p1 and p4 as the initial centroids.

(40%)